

Database Systems - MCQ Question Bank (Set 5)

20 Multiple Choice Questions | Includes 2 repeated from earlier sets | Answer Key on Last Page

- 1. Which SQL clause is used to filter groups after aggregation?**
 - A. WHERE
 - B. GROUP BY
 - C. HAVING
 - D. ORDER BY
- 2. Which of the following best describes a weak entity in an ER diagram?**
 - A. Has its own primary key
 - B. Cannot be uniquely identified without a related strong entity
 - C. Has no attributes
 - D. Cannot participate in any relationship
- 3. Which of the following best describes a natural join?**
 - A. A join that requires an explicit ON condition
 - B. A join that automatically matches columns with the same name in both tables
 - C. A join that always returns an empty result
 - D. A join used only for self-referencing tables
- 4. Which of the following is a valid use of the SQL CASE expression?**
 - A. To define a new table
 - B. To perform conditional logic within a query
 - C. To grant user privileges
 - D. To create an index
- 5. In a database, an attribute domain refers to:**
 - A. The set of allowable values for an attribute
 - B. The physical storage location of a table
 - C. The number of rows in a table
 - D. The name of the database server
- 6. Which of the following best describes eventual consistency in distributed databases?**
 - A. All replicas are updated instantly and always consistent
 - B. Replicas will become consistent over time if no new updates occur
 - C. Data is never replicated
 - D. Only one replica is allowed to exist
- 7. Which SQL command is used to permanently save changes made during a transaction?**
 - A. SAVE
 - B. COMMIT
 - C. FLUSH
 - D. APPLY
- 8. Which SQL command reverses changes made in the current uncommitted transaction?**
 - A. ROLLBACK
 - B. UNDO

- C. RESET
- D. CANCEL

9. Which of the following best describes a clustered index?

- A. An index stored separately from the table with no effect on row order
- B. An index that determines the physical order of data rows in a table
- C. An index that can only be created on primary keys
- D. An index used only for text columns

10. Which of the following best describes database sharding versus replication?

- A. Sharding duplicates all data on every node, replication splits data
- B. Sharding splits data across nodes, replication duplicates data across nodes
- C. Both terms mean the same thing
- D. Neither applies to distributed databases

11. Which of the following is true about a many-to-many relationship in a relational database?

- A. It can be directly represented without any additional table
- B. It requires a junction (bridge) table to be implemented
- C. It is not allowed in relational databases
- D. It always results in a NULL foreign key

12. Which of the following best describes the purpose of the SQL ALTER TABLE statement?

- A. To insert new rows into a table
- B. To modify the structure of an existing table
- C. To delete a table
- D. To create a backup of a table

13. Which of the following best describes a dirty read?

- A. Reading data that has been permanently deleted
- B. Reading uncommitted data from another transaction that may later be rolled back
- C. Reading data from an encrypted column
- D. Reading data faster than normal

14. Which of the following best describes phantom reads?

- A. Reading a row that no longer exists in memory
- B. A transaction re-executes a query and finds new rows inserted by another transaction
- C. Reading corrupted data from disk
- D. Reading data without proper authentication

15. Which of the following best describes a data warehouse?

- A. A system optimized for fast day-to-day transactional updates
- B. A centralized repository optimized for analysis and reporting of historical data
- C. A temporary cache for session data
- D. A single table used to store user passwords

16. Which of the following best describes the difference between DELETE and TRUNCATE in SQL?

- A. DELETE can use a WHERE clause and is logged row by row; TRUNCATE removes all rows and is minimally logged

- B. TRUNCATE can use a WHERE clause but DELETE cannot
- C. They are functionally identical in all databases
- D. DELETE removes the table structure, TRUNCATE does not

17. Which of the following best describes database replication?

- A. Splitting a table into smaller pieces
- B. Maintaining copies of the same data on multiple servers for availability and fault tolerance
- C. Encrypting data before storage
- D. Removing duplicate rows from a table

18. Which of the following best describes a primary key constraint's implicit behavior?

- A. It allows multiple NULL values
- B. It automatically creates a unique index and disallows NULL values
- C. It disables indexing on the column
- D. It only applies to numeric columns

19. Which of the following best describes the term 'query optimizer' in a DBMS?

- A. A tool that backs up the database
- B. A component that selects the most efficient execution strategy for a query
- C. A tool that formats query output for display
- D. A component that manages user authentication

20. Which of the following is an advantage of using indexes in a database?

- A. They always reduce storage space
- B. They speed up data retrieval at the cost of additional storage and slower writes
- C. They eliminate the need for primary keys
- D. They guarantee data is never duplicated

Answer Key

1. C 2. B 3. B 4. B

5. A 6. B 7. B 8. A

9. B 10. B 11. B 12. B

13. B 14. B 15. B 16. A

17. B 18. B 19. B 20. B